Math 210B Notes

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Chapter 1

Introduction to Rings

1.1 Ring and Field Definitions

We move on from studying groups to studying rings and fields. First, lets compare some analogues between groups and rings.

Groups:

(i) 1 operation
(ii) Subgroups
(iii) Normal groups N
(iv) Quotient groups G/N
(v) Morphisms of groups

Rings:

(i) 2 operations
(ii) Subrings
(iii) Ideals I
(iv) Quotient rings R/I
(v) Morphisms of rings

We build the theory of rings and fields in a similar way to the theory of groups. An important type of ring we wish to study is the ring of polynomials with coefficients in a field. Our goal is to be able to study Galois Theory and make a connection between automorphisms of fields and their subfields.

Before we get started, consider the following example.

Example 1.1.1. Let R be a set with operations $+$ and $\times$ such that distribution holds for all elements in the set:

$$\begin{aligned} \forall a, b, c \in R \quad a \times (b + c) &= (a \times b) + (a \times c) \\ (a + b) \times c &= (a \times c) + (b \times c) \end{aligned}$$

Further assume $+$ and $\times$ are associative and that there exists $1 \in R$ such that $1 \times a = a \times 1 = a$ for all $a \in R$. Let $-a$ and $-b$ be the additive inverses of $a, b \in R$. Show that

$$a + b = b + a$$

Proof.

$$\begin{aligned} (a + b) \times (1 + 1) &= (a + b) \times 1 + (a + b) \times 1 \\ &= a \times 1 + b \times 1 + a \times 1 + b \times 1 \\ &= a + b + a + b \end{aligned} \tag{I}$$

$$\begin{aligned} (a + b) \times (1 + 1) &= a \times (1 + 1) + b \times (1 + 1) \\ &= a \times 1 + a \times 1 + b \times 1 + b \times 1 \\ &= a + a + b + b \end{aligned} \tag{II}$$

From (I) and (II), we have

$$\begin{aligned} a+b+a+b &= a+a+b+b \\ \implies a+b+a+b-b &= a+a+b+b-b \\ \implies -a+a+b+a+0 &= -a+a+a+b+0 \\ \implies 0+b+a &= 0+a+b \\ \implies b+a &= a+b \end{aligned}$$

□

This example motivates the following definition.

Definition 1.1.1. (Ring)
A ring is a set R together with two binary operations $+$ (called addition) and $\times$ (called multiplication) satisfying the following:

(i) $(R,+)$ is an abelian group

(ii) Multiplication is associative $\forall a,b,c \in R$

$$(a \times b) \times c = a \times (b \times c)$$

(iii) Distributive laws hold $\forall a,b,c \in R$

$$\begin{aligned} \text{Left distribution: } a \times (b \times c) &= (a \times b) + (a \times c) \\ \text{Right distribution: } (a+b) \times c &= (a \times c) + (b \times c) \end{aligned}$$

If multiplication is commutative, we call R a commutative ring. The ring R is said to have an identity denoted 1 (or contains a unity element) if

$$1 \times a = a \times 1 = a \quad \forall a \in R$$

In this case, R is called a ring with unity.

Notation . $a \times b$ will be written as ab. The additive identity of $(R,+)$ will be denoted 0. The additive inverse of an element $a \in R$ will be denoted $-a$.

Notice that our definition for a ring does not require the existence of a multiplicative inverse for each element in the ring. The addition of multiplicative inverses leads to more specific types of rings, and with the addition of multiplicative commutativity, we get fields.

Definition 1.1.2. (Division Ring, Field)
A ring R with unity 1 (where $1 \neq 0$) is called a division ring if every $a \in R$ where $a \neq 0$ has an element $b \in R$ such that $ab = ba = 1$. That is, if all nonzero elements have a multiplicative inverse. If R is also commutative, then R is called a field.

Example 1.1.2. (i) Trivial rings: Given any group $(G,*)$ if we take $*$ as addition and define multiplication as $ab = 0 \quad \forall a,b \in G$, then this forms a ring.

(ii) If $R = \{0\}$, this is called the zero ring with multiplication and addition defined as $0 \cdot 0 = 0$ and $0+0=0$. Note that this is the only ring where $1=0$. Show that if $1=0$, then $R=\{0\}$.

Proof. Let $a \in R$.

$$\begin{aligned} a \cdot 0 = a(0+0) &= a \cdot 0 + a \cdot 0 \\ \implies a \cdot 0 &= a \cdot 0 + a \cdot 0 \\ \implies 0 = a \cdot 0 &= a \cdot 1 = a \end{aligned}$$

□

Many theorems will state $1 \neq 0$ instead of $R \neq 0$.

(iii) $\mathbb{Z}$ with the usual multiplication and addition. Note that in $\mathbb{Z}/\{0\}$ we do not have a group with

respect to multiplication.

(iv) $\mathbb{Q}$ is a ring with the usual operations and $\mathbb{Q}/\{0\}$ is a group with respect to multiplication, that is $\mathbb{Q}$ is a field (multiplication in $\mathbb{Q}$ is commutative). $\mathbb{C}$ and $\mathbb{R}$ are fields as well.

(v) $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring with unity $\bar{1}$ ($\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, ..., n \bar{-} 1\}$) where the multiplication is defined $\bar{a} \cdot \bar{b} = \bar{ab}$.

(vi) The quaternians: recall the imaginary units $i^2 = j^2 = k^2 = ijk = -1$. Looking at the set $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ where addition is defined by

$$(a + bi + cj + dk) + (a' + b'i + c'j + d'k) = (a + a') + (b + b')i + (c + c')j + (d + d')k$$

and multiplication is defined by distribution

$$\begin{aligned}&(a + bi + cj + dk)(a' + b'i + c'j + d'k)\\ =&aa' - bb' - cc' - dd' + (ab' + ba' + cd' - dc')i + (a'c - bd' + ca' + db')j + (ad' + bc' - cb' + da')k\end{aligned}$$

Then $\mathbb{H}$ forms a ring. We see that, for $x \in \mathbb{H}$,

$$\begin{aligned}x\bar{x} &= (a + bi + cj + dk)(a - bi - cj - dk) = a^2 + b^2 + c^2 + d^2\\ x^{-1} &= \frac{\bar{x}}{x\bar{x}} = \frac{a - bi - cj - dk}{a^2 + b^2 + c^2 + d^2}\end{aligned}$$

each analogous to the complex numbers. Every $x \neq 0$ in $\mathbb{H}$ has a multiplicative inverse. However, multiplication does not commute in all of $\mathbb{H}$ ($ik = -j \neq j = ki$), so $\mathbb{H}$ is a division ring but not a field.

(vii) Let X be a nonempty set and A be any ring. The set of all mappings $f : X \to A$ where $(f + g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$ forms a ring.

Since rings add an additional operation to a group structure, we have new properties that arise from the interaction of the two operations.

Proposition 1.1.1. Let R be a ring.

(i) $0 \cdot a = a \cdot 0 = 0 \quad \forall a, b \in R$

(ii) $(-a)b = a(-b) = -(ab) \quad \forall a, b \in R$

(iii) $(-a)(-b) = ab \quad \forall a, b \in R$

(iv) If R has identity 1, then that identity is unique and

$$-a = (-1)a \quad \forall a \in R$$

Proof. (i) Given $a \in R$, we have

$$\begin{aligned}a \cdot 0 = a(0 + 0) &= a \cdot 0 + a \cdot 0\\ \implies 0 &= a \cdot 0\end{aligned}$$

Similarly,

$$\begin{aligned}0 \cdot a = (0 + 0)a &= 0 \cdot a + 0 \cdot a\\ \implies 0 &= 0 \cdot a\end{aligned}$$

(ii) Given $a, b \in R$, we have

$$\begin{aligned}ab + (-a)b &= (a + -a)b\\ &= 0 \cdot b\\ &= 0 \implies -(ab) \qquad = (-a)b\end{aligned}$$

$-(ab) = a(-b)$ is analogous.

(iii) Given $a, b \in R$, we have

$$\begin{aligned} -(ab) + (-a)(-b) &= (-a)b + (-a)(-b) \\ &= (-a)(b + -b) \\ &= (-a) \cdot 0 \\ &= 0 \\ \implies -(-(ab)) &= (-a)(-b) \\ \implies ab &= (-a)(-b) \end{aligned}$$

(iv) Let 1 and e both be identity elements in R. Then

$$\left.\begin{aligned} 1 \cdot e &= e \\ 1 \cdot e &= 1 \end{aligned}\right\} \implies 1 = e$$

Thus the identity element of R is unique. Let $a \in R$ be given. We have

$$\begin{aligned} 0 &= (1 + (-1))a \\ &= 1 \cdot a + (-1) \cdot a \\ &= a + (-1)a \\ \implies -a &= (-1)a \end{aligned}$$

□

1.2 Zero Divisors and Integral Domains

This semester we are adding multiplicative structure and we hope it is a good structure, but sometimes it is not. Consider the statement $3 \cdot 4 = 0$. Is this a true statement? In $\mathbb{R}$, no, it is not. However, if we consider the ring $\mathbb{Z}/6\mathbb{Z}$, then the statement is true. We call this blend of bad multiplicative behavior zero divisors.

Definition 1.2.1. (Zero Divisor)
Let R be a ring. A nonzero element $a \in R$ is called a zero divisor if there exists a nonzero $b \in R$ such that $ab = 0$ or $ba = 0$.

Example 1.2.1. Consider the matrices in $M_2(\mathbb{R})$. Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$. Then

$$AB = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$
$$BA = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

We now introduce the notion of multiplicative inverses. For a ring to be field, every element needs to have a multiplicative inverse. However, in some rings we might have some elements that have inverses but not all. We call these elements units.

Definition 1.2.2. (Unit)
Assume a ring R has identity $1 \neq 0$. An element $a \in R$ is called a unit in R if there exists an element $u \in R$ such that $uv = 1 = vu$. The set of all units in R is denoted by $R^\times$.

Example 1.2.2. $\mathbb{Z}/6\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$, $\mathbb{Z}/9\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}, \bar{7}, \bar{8}\}$.
Here, $(\mathbb{Z}/6\mathbb{Z})^\times = \{\bar{1}, \bar{5}\}$ and $(\mathbb{Z}/9\mathbb{Z})^\times = \{\bar{1}, \bar{2}, \bar{4}, \bar{5}, \bar{7}, \bar{8}\}$.

Concerning rings of the form $\mathbb{Z}/n\mathbb{Z}$ for $n \geq 0$, we have the following result regarding units.

Proposition 1.2.1. $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ is a unit if and only if $\gcd(a, n) = 1$.

Proof. ($\Longrightarrow$) Suppose $\bar{a}$ is a unit. Then there exists $\bar{b} \in R$ such that $\bar{a}\bar{b} = \bar{1}$. This means that $ab \equiv 1($ mod $n)$, so $n \mid ab - 1$. Thus there exists some $y \in \mathbb{Z}$ such that $ab - 1 = ny$. More specifically, $ab - ny = 1$. Thus any common divisor d of a and n must also divide 1, so $d = 1$. Hence $\gcd(a, n) = 1$.
($\Longleftarrow$) Suppose $\gcd(a, n) = 1$. By Bézout's identity, there exist $x, y \in \mathbb{Z}$ such that $ax + ny = 1$. Thus $ax \equiv 1($ mod $n)$, so $\bar{a}\bar{x} = \bar{1}$. Hence, $\bar{a}$ is a unit. □

Remark 1.2.1. A consequence of the above proposition: $\mathbb{Z}/p\mathbb{Z}$ where p is prime is a field as $(\mathbb{Z}/p\mathbb{Z})^t \, imes = \{\bar{1}, \bar{2}, ..., p \bar{-} 1\}$.

Remark 1.2.2. Any nonzero element $\bar{a}$ of $\mathbb{Z}/n\mathbb{Z}$ with $\gcd(a, n) > 1$ is a zero divisor.

Example 1.2.3. In $Z/12\mathbb{Z}$, $\bar{8}$ is not relatively prime to 12 and $\bar{8}\bar{3} = \bar{2}\bar{4} = \bar{0}$.

In general, given $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ with $\bar{a} \neq 0$, we have that

$$\bar{a}\overline{\left(\frac{n}{\gcd(a, n)}\right)} = \frac{\bar{a}}{\gcd(a, n)} \cdot \bar{n} = \bar{0}$$

as $\frac{a}{\gcd(a,n)} \in \mathbb{Z}$. Further, since $\gcd(a, n) > 1$, we have that

$$0 < \frac{n}{\gcd(a, n)} < n$$

Proposition 1.2.2. Zero divisors can never be units.

Proof. Let R be a ring. Suppose that $a \in R$ is a unit and a zero divisor. Since a is a unit,

$$\exists b \in R \text{ such that } ab = 1 = ba$$

Since a is a zero divisor, "

$$\exists c \in R \text{ such that } c \neq 0 \text{ and } ac = 0 \text{ or } ca = 0$$

If $ac = 0$, then

$$\begin{aligned} b(ac) &= b \cdot 0 \\ \implies (ba)c &= 0 \\ \implies 1 \cdot c &= 0 \\ \implies c &= 0 \end{aligned}$$

a contradiction. The case for $ca = 0$ is analogous. □

It follows that fields contain no zero divisors. We now introduce the notion of an integral domain which is a ring with no zero divisors.

Definition 1.2.3. (Integral Domain)
A commutative ring with identity $1 \neq 0$ si called an integral domain if it has no zero divisors.

Proposition 1.2.3. Let R be a ring and assume $a, b, c \in R$ with a not a zero divisor. If $ab = ac$, then $b = c$ or $a = 0$.

Proof.

$$\begin{aligned} ab &= ac \\ \implies ab - ac &= 0 \\ \implies a(b - c) &= 0 \end{aligned}$$

a is not a zero divisor, so either $a = 0$ or $a \neq 0$. If $a = 0$, then we are done. If $a \neq 0$, then $b - c = 0$ which implies $b = c$. □

The proposition above applies for any integral domain. From this proposition, we can derive the following corollary.

Corollary 1.2.1. Any finite integral domain is a field.

Proof. Let R be a finite integral domain and let $a \in R$ such that $a \neq 0$. Define the mapping

$$\begin{aligned} f_a &: R \to R \\ f_a(x) &= ax \quad \forall x \in R \end{aligned}$$

By the previous proposition, f_a is injective and since R is finite f_a is a bijection. In particular,

$$\exists b \in R \text{ such that } f_a(b) = ab = 1$$

Therefore a is a unit and since a was an arbitrary nonzero element of R, R is a field. □

In group theory, an important structure of study was the subgroup. For rings, we have an analogous structure called a subring.

Definition 1.2.4. (Subring)
A subring of a ring R is a subgroup of R with respect to addition which is closed under multiplication.

Proposition 1.2.4. (Subring Check)
Let $H \subseteq R$ where R is a ring.

(i) Subgroup check:

  (a) Show H is nonempty (e.g., show $0 \in H$)

  (b) Show $\forall x, y \in H \quad x - y \in H$

(ii) Closure under multiplication: $\forall x, y \in H \quad xy \in H$

1.3 Polynomials and Ring Homomorphisms

One of the big motivations for studying rings is to understand polynomials. Polynomials are a fundamental object in algebra and in this section we will define ring homomorphisms to help us understand the structure of polynomials.

Definition 1.3.1. (Polynomial)
Let R be a ring with unity 1. Let x be an indeterminate.

(i) The sum
$$\sum_{i=0}^{n} a_i x^i = a_0 + a_1 x + a_2 x^2 + ... + a_n x^n = f(x)$$
with $n \geq 0$ and each $a_i \in R$ is called a polynomial in x with coefficients in R.

(ii) If $a_n \neq 0$, then $\deg(f(x)) = n$. The zero polynomial has all $a_i = 0$, and the degree of the zero polynomial is undefined. The zero polynomial is denoted by 0.

(iii) The collection of all polynomials with coefficients in R is denoted $R[x]$, and is a ring with respect to the opperations
$$\left(\sum_{i=0}^{n} a_i x^i\right) + \left(\sum_{i=0}^{n} b_i x^i\right) = \sum_{i=0}^{n} (a_i + b_i) x^i$$
and
$$\left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{i=0}^{n} b_i x^i\right) = a_0 b_0 + (a_0 b_1 + b_0 a_1) x + (a_0 b_2 + a_1 b_1 + a_2 b_0) x^2 + ... + (a_0 b_n + a_1 b_{n-1} + ... + a_n b_0) x^n$$

Example 1.3.1. When $R = \mathbb{Z}/3\mathbb{Z}$ (the polynomial ring $\mathbb{Z}/3\mathbb{Z}[x]$) Let $p(x) = x^2 + 2x + 1$ and $q(x) = x^3 + x + 2$. Then
$$p(x) + q(x) = x^3 + x^2 + 3x + 3 = x^3 + x^2$$
and
$$p(x)q(x) = x^5 + 2x^4 + 2x^3 + 4x^2 + 5x + 2$$

Example 1.3.2. In $\mathbb{Z}/2\mathbb{Z}[x]$
$$x^2 + 1 = (x + 1)(x + 1)$$

Ring homomorphisms are defined in a similar way to group homomorphisms.

Definition 1.3.2. (Ring Homomorphism)
Let R and S be rings.

(i) A ring homormorphism is a mapping $\phi : R \to S$ such that for all $a, b \in R$

(a) $\phi(a + b) = \phi(a) + \phi(b)$

(b) $\phi(ab) = \phi(a)\phi(b)$

(ii) The kernel of ϕ is defined as
$$\ker \phi = \{x \in R : \phi(x) = 0_S\}$$

(iii) A bijective ring homomorphism is an isomorphism.

Example 1.3.3. Let $\phi : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ be defined by
$$\phi(x) = \begin{cases} 0 & \text{if } x \text{ is even} \\ 1 & \text{if } x \text{ is odd} \end{cases}$$

Case 1: x and y are both even.
$$\phi(x + y) = 0 = \phi(x) + \phi(y), \phi(xy) = 0 = 0 \cdot 0 = \phi(x)\phi(y)$$

Case 2: x and y are both odd.
$$\phi(x + y) = 0 = 1 + 1 = \phi(x) + \phi(y), \phi(xy) = 1 = 1 \cdot 1 = \phi(x)\phi(y)$$

Case 3: One of x and y is even and the other is odd.
Without loss of generality, let x be even and y be odd.

$$\phi(x+y) = 1 = 0 + 1 = \phi(x) + \phi(y), \phi(xy) = 0 = 0 \cdot 1 = \phi(x)\phi(y)$$

We have two notable properties involving ring homomorphisms.

Proposition 1.3.1. Let R and S be rings and let $\phi : R \to S$ be a ring homomorphism.

(i) $\phi(R)$ is a subring of S.

(ii) $\ker \phi$ is a subring of R. Furthermore, if $\alpha \in \ker \phi$ and $r \in R$, the $r\alpha \in \ker \phi$ and $\alpha r \in \ker \phi$. That is, $\ker \phi$ is closed under the entire ring multiplication.

Proof. (i) We know that $\phi(R)$ is an additive subgroup of S from last semester. It remains to show that if $s_1, s_2 \in \phi(R)$, then $s_1 s_2 \in \phi(R)$. Let $s_1, s_2 \in \phi(R)$ be given. Then

$$\exists r_1, r_2 \in R \text{ such that } s_1 = \phi(r_1) \text{ and } s_2 = \phi(r_2)$$

Then

$$s_1 s_2 = \phi(r_1)\phi(r_2) = \phi(r_1 r_2) \in \phi(R)$$

as desired.

(ii) We know that $\ker \phi$ is an additive subgroup of R from last semester. It suffices to show that if $r \in R$ and $\alpha \in \ker \phi$, then $r\alpha, \alpha r \in \ker \phi$. Let $r \in R$ and $\alpha \in \ker \phi$ be given. We have

$$\begin{aligned} \phi(r\alpha) &= \phi(r)\phi(\alpha) \\ &= \phi(r) \cdot 0_S \\ &= 0_S \\ \implies r\alpha &\in \ker \phi \end{aligned}$$

Similarly,

$$\begin{aligned} \phi(\alpha r) &= \phi(\alpha)\phi(r) \\ &= 0_S \cdot \phi(r) \\ &= 0_S \\ \implies \alpha r &\in \ker \phi \end{aligned}$$

□

The kernel of a ring homomorphism is a special type of subring. In particular, the kernel of a ring homomorphism is closed under multiplication by any element in the ring. We call this the absorbative property it allows us to define coset multiplication for rings.

Let I be the kernel of a ring homomorphism $\phi : R \to S$. We know that the additive left cosets of I form a group. Because every subgroup of an abelian group is normal, the definition of coset addition is well-defined:

$$(\alpha + I) + (\beta + I) = (\alpha + \beta) + I \quad \forall \alpha, \beta \in R$$

We aim to show that the absorbative property of $\ker \phi = I$ makes coset multiplication well-defined. We define coset multiplication as

$$(\alpha + I)(\beta + I) = (\alpha\beta) + I$$

Suppose that $\alpha + I = \alpha' + I$ and $\beta + I = \beta' + I$. We want to show that

$$(\alpha\beta) + I = (\alpha'\beta') + I$$

Note that

$$\begin{aligned} \alpha + I = \alpha' + I &\implies \alpha = \alpha' + x \text{ where } x \in I \\ \beta + I = \beta' + I &\implies \beta = \beta' + y \text{ where } y \in I \end{aligned}$$

Then

$$\alpha\beta = (\alpha' + x)(\beta' + y) = \alpha'\beta' + \alpha' y + x\beta' + xy$$

$\alpha' y, x\beta', xy \in I$, so $\alpha\beta = \alpha'\beta' + z$ where $z = \alpha' y + x\beta' + xy \in I$. Thus $\alpha\beta - \alpha'\beta' \in I$. Therefore, $(\alpha\beta) + I = (\alpha'\beta') + I$ and so coset multiplication is well-defined. We have now motivated the following definition.

Definition 1.3.3. (Ideals)
Let R be a ring and let $I \subseteq R$. Let $r \in R$.

(i) $rI = \{rx : x \in I\}$ and $Ir = \{xr : x \in I\}$

(ii) I is a left ideal of R if

  (a) I is a subring of R

  (b) $\forall r \in R \;\; rI \subseteq I$

(iii) I is a right ideal of R if

  (a) I is a subring of R

  (b) $\forall r \in R \;\; Ir \subseteq I$

(iv) If I is both a left and right ideal of R, then I is called an ideal of R.

1.4 Quotient Rings and the Isomorphism Theorems

Now that we have defined ideals, we can use them to construct new rings called quotient rings, similar to how we used normal subgroups to construct quotient groups. We will then use ideals and quotient rings to discuss isomorphisms between ring structures. First, given a ring R and an ideal I of R, we will show that the set R/I has any meaning.

Proposition 1.4.1. Let R be a ring and I be an ideal of R. Then R/I is a ring under the binary operations

$$\forall r, s \in I \quad (r+I)+(s+I) = (r+s)+I$$
$$\forall r, s \in I \quad (r+I)(s+I) = (rs)+I$$

Proof. Note that we proved multiplication is well-defined and we also know the group structure exists from last semester. Distribution and associativity in R/I follow from distribution and associativity in R. □

Definition 1.4.1. (Quotient Ring)
When I is an ideal of a ring R, the ring R/I is called the quotient ring of R by I.

We can now establish the isormophism theorems for rings.

Theorem 1.4.1. (The First Isomorphism Theorem for Rings)

(i) If $\phi : R \to S$ is a ring homomorphism, then $\ker\phi$ is an ideal of R, $\phi(R)$ is a subring of S, and

$$\frac{R}{\ker\phi} \cong \phi(R)$$

(ii) If I is any ideal of R, then the map

$$\pi : R \to R/I$$
$$\pi(r) = r + I$$

is a surjective ring homomorphism with kernel I. This is called the natural projection of R onto R/I.

Proof. Last semester we established the mapping

$$\mu : R/\ker\phi \to \phi(R)$$
$$\mu(r+K) = \phi(r)$$

as a well-defined bijection which preserved the group operation (in this case, $+$). It remains to show that μ preserves coset multiplication.

$$\begin{aligned}\mu\left((r+K)(s+K)\right) &= \mu(rs+K)\\ &= \phi(rs)\\ &= \phi(r)\phi(s)\\ &= \mu(r+K)\mu(s+K)\end{aligned}$$

□

Example 1.4.1. Let R be a ring.

(i) The trivial ideal is 0.

(ii) R is an ideal of R.

(iii) If I is an ideal of R where $I \neq R$, then I is a proper ideal of R.

Example 1.4.2. Last semester (when studying cyclic groups) we proved the only subgroups of $\mathbb{Z}$ were of the form $n\mathbb{Z}$ for some $n \geq 0$. This tells us that all the ideals of $\mathbb{Z}$ must be of the same form. It is a

straightforward exercise to show that $n\mathbb{Z}$ $(n \geq 0)$ is an ideal of $\mathbb{Z}$:

$$x \in \mathbb{Z}, y \in n\mathbb{Z} \implies y = nz \quad \text{for } z \in \mathbb{Z} \text{ and } xy = x(nz) = n(xz) \in n\mathbb{Z}$$
$$yx = (nz)x = n(zx) \in n\mathbb{Z}$$

In this setting,

$$\begin{aligned} &\pi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \\ &\pi(x) = \bar{x} \text{ (reduction modulo } n) \\ &\mathbb{Z}/5\mathbb{Z} \quad 7 = 2 \end{aligned}$$

Example 1.4.3. Consider $I = \{p(x) \in \mathbb{Z}[x] : p(x) = 0 \text{ or whose terms are of at least degree } 2\}$. Does I have the absorbative property of an ideal? Let $f(x) \in \mathbb{Z}[x]$ and $q(x) \in I$. Then

$$f(x) = \sum_{i=0}^{n} a_i x^i \quad q(x) = \sum_{i=0}^{n} b_i x^i \quad \text{where } b_0 = 0, b_1 = 0$$

Then the constant coefficient of $f(x)q(x)$ is $a_0 b_0 = a_0 \cdot 0 = 0$ and the linear coefficient of $f(x)q(x)$ is $a_1 b_0 + a_0 b_1 = a_1 \cdot 0 + a_0 \cdot 0 = 0$.

Example 1.4.4. Let A be a ring and X be a nonempty set. Let R be the ring of all functions from X to A. For each $c \in X$ we can define the mapping

$$E_c : R \to A$$
$$E_c(f) = f(c) \text{ (the evaluation homomorphism)}$$

Let $f, g \in R$. Then

$$\begin{aligned} E_c\left((f \cdot g)(x)\right) &= (f \cdot g)(c) \\ &= f(c) \cdot g(c) \\ &= E_c\left(f(x)\right) \cdot E_c\left(g(x)\right) \end{aligned}$$

Addition follows similarly. Notice

$$\ker E_c = \{f \in R : E_c(f) = 0\} = \{f \in R : f(c) = 0\}$$

By the first isomorphism theorem, we have

$$\frac{R}{\ker E_c} \cong E_c(R)$$

What is $E_c(R)$? Let $a \in A$. Notice the constant function $h(x) = a \ \forall x \in X$ gives

$$E_c\left(h(x)\right) = h(c) = a$$

Thus $E_c(R)$ is surjective and so

$$\frac{R}{\ker E_c} \cong A$$

Theorem 1.4.2. (The Second, Third, and Fourth Isomorphism Theorems for Rings)
Let R be a ring.

(i) Let A be a subring of R and let B be an ideal of R. Then

$$A + B = \{a + b : a \in A, b \in B\}$$

is a subring of R, $A \cap B$ is an ideal of A, and

$$\frac{A + B}{B} \cong \frac{A}{A \cap B}$$

(ii) Let I and J be ideals of R with $I \subseteq J$. Then J/I is an ideal of R/I and

$$\frac{R/I}{J/I} \cong \frac{R}{J}$$

(iii) Let I be an ideal of R. The correpsondence $A \leftrightarrow A/I$ is an inclusion preserving bijection between the set of subrings of R containing I and the subrings of R/I. Furthermore, A (a subring containing I) is an ideal of R if and only if A/I is an ideal of R/I.

Similar to groups, ring homomorphisms will map subrings to subrings. Can the same be said for ideals? The answer is no, this is only true for surjective homomorphisms.

1.5 Generated Ideals

Proposition 1.5.1. Let I and J be ideals of a ring R. Define

$$I + J = \{a + b : a \in I, b \in J\}$$

Then $I + J$ is the smalles ideal of R that contains both I and J.

Proof. We know that I and J are normal additive subgroups of R, so $I + J$ is a subgroup of R.

$$\begin{aligned} a_1 + b_1, a_2 + b_2 \in I + J \implies & (a_1 + b_1)(a_2 + b_2) \\ & = a_1a_2 + a_1b_2b_1a_2 + b_1b_2 \end{aligned}$$

Notice

$$\begin{aligned} & a_1a_2 \in I \\ a_1 \in I \implies & a_1b_2 \in I \end{aligned} \qquad \begin{aligned} & b_1b_2 \in J \\ b_1 \in J \implies & b_1a_2 \in J \end{aligned}$$

since I and J are ideals. Hence, $(a_1 + b_1)(a_2 + b_2) \in I + J$ and so $I + J$ is an ideal of R.
Futhermore, $I + J$ is the smalles ideal of R which contains I and J. We can verify that I and J are contained in $I + J$. Letting $b = 0$, we see any $a \in I$ is in $I + J$. Similarly, letting $a = 0$ we see any $b \in J$ is in $I + J$. Hence both I and J are contained in $I + J$. Let K be an ideal of R which contains both I and J. Let $a + b \in I + J$. Then

$$\begin{aligned} a \in I \text{ and } b \in J & \implies a \in K \text{ and } b \in K \\ & \implies a + b \in K \\ & \implies I + J \subseteq K \end{aligned}$$

□

Definition 1.5.1. (Finitely Generated Ideal)
Let R be a ring with identity 1. Let A be a subset of R. Let (A) denote the smallest ideal of R containing A. We can define (A) more concretely by

$$(A) = \bigcap_{I \subseteq A} I, \ I \text{ is an ideal of } R$$

An ideal genearted by a finite set is called a finitely generated ideal. If $A = \{a_1, a_2, ..., a_n\}$, then (A) will be written as $(a_1, a_2, ..., a_n)$.
$(I, J) = I \cup J$ (where I and J are subsets of R)

What is the simplest group generated by a set? Cyclic groups that are generated with a single element. Given a ring R with unity $1 \neq 0$ and some $a \in R$, consider (a), the ideal generated by a. (a) needs to contain ra and ar for all $r \in R$. What's more is (a) must be closed with respect to addition so we need all finite sums of ra's and ar's

$$(r_1a + r_2a + ar_3 + ar_4 + ...)$$

If we suppose that R is commutative, then (a) is very simple.

Definition 1.5.2. (Principal Ideal)
Let R be a ring and $a \in R$. When R is commutative with $1 \neq 0$,

$$(a) = \{ar : r \in R\}$$

and (a) is called the principal ideal of R generated by a.

Example 1.5.1. (i) In any ring, $0 = (0)$ and when R has unity 1, $(1) = R$.

(ii) Let R be a commutatie ring with unity $1 \neq 0$.

$$(a) + (b) = (a + b) \text{ (from what we proved about } I + J)$$

(iii) All ideals are principal.

(iv) All additive subgroups of $\mathbb{Z}$ are of the form $n\mathbb{Z}, n \geq 0$ and $n\mathbb{Z}$ is an ideal for all $n \geq 0$. Furthermore, $n\mathbb{Z} = \{nx : x \in \mathbb{Z}\} = (n)$.

Proposition 1.5.2. Let I be an ideal of R.

(i) $I = R$ if and only if I contains a unit.

(ii) Assume R is commutative. R is a field if and only if its only ideals are 0 and R.

Proof. (i) If $I = R$, then $1 \in I$. Conversely, if u is a unit in I with inverse $v \in R$, then for any $r \in R$

$$r = r \cdot 1 = r(vu) = (rv)u \in I$$

Thus $R = I$.

(ii) The ring R is a field if and only if every nonzero element is a unit. Hence, any nonzero ideal contains a unit. Thus the only nonzero ideal of R is R. Conversely, if 0 and R are the only ideals of R, we let u be any nonzero element of R. By hypothesis, $(u) = R$ so $1 \in (u)$. That is, $1 = uv$ for some $v \in R$. That is, u is a unit. Thus every nonzero element of R is a unit so R is a field.

□

Let's say I is a nonzero ideal in $\mathbb{Q}$. Then there exists $a \in I$ such that $a \neq 0$. Since ideals have the absorbative property, we know that $\frac{1}{a} \in \mathbb{Q}$ and $\frac{1}{a} \cdot a \in I$ which implies $1 \in I$. Then for any $b \in \mathbb{Q}$ we have $b = b \cdot 1 \in I$ which implies $\mathbb{Q} \subseteq I$. Thus $I = \mathbb{Q}$.

1.6 Maximal Ideals

Note that for any $n \in \mathbb{N}$ and $a \in R$ a ring, we define

$$na = a + a + ... + a \quad n \text{ times}$$

Proposition 1.6.1. Let $m, n \in \mathbb{N}$ such that $\gcd(m, n) = 1$. Then $m \mid \binom{m}{n}$.

Proof.

$$\begin{aligned}\binom{m}{n} &= \frac{m!}{n!(m-n)!} \\ &= \frac{m}{n}\frac{(m-1)!}{(n-1)!(m-n)!} \\ &= \frac{m}{n}\binom{m-1}{n-1}\end{aligned}$$

So,

$$n\binom{m}{n} = m\binom{m-1}{n-1}$$

so $m \mid n\binom{m}{n}$. But $\gcd(m, n) = 1$, so $m \mid \binom{m}{n}$. □

Corollary 1.6.1. If R is a field, then any nonzero ring homomorphism from R into another ring is an injection.

Proof. Let R be a field and let $\phi : R \to S$ be a nonzero ring homomorphism. Since $\ker \phi$ is an ideal fo the field R, $\ker \phi$ is either 0 or R itself. If $\ker \phi = R$, then $\phi(r) = 0$ for all $r \in R$. Hence, contrary to our assumption, ϕ is the zero homomorphism. This leaves us with $\ker \phi = 0$. As we know from our work with groups,

$$\ker \phi = 0 \iff \phi \text{ is injective}$$

□

Definition 1.6.1. (Maximal Ideal)
An ideal M in a ring S is called a maximal ideal if $M \neq S$ ans the only ideals that contain M are S and M itself.

A maximal ideal gives the notion of a "largest proper ideal" of a ring. That is, if M is the maximal ideal to a ring R and H is a proper ideal of R such that $M \subseteq H \subseteq R$, then either $H = R$ or $H = M$.

Proposition 1.6.2. In a ring with unity, every proper ideal is contained in a maximal ideal.

The next result characterizes maximal ideals with their quotient structures of a commutative ring.

Proposition 1.6.3. Assume R is commutative. The ideal M is a maximal ideal if and only if R/M is a field.

Proof. M is a maximal ideal if and only if there are no ideals I with

$$M \subset I \subset R$$

By the fourth isomorphism theorem, ideals of R containing M are in 1-1 correspondence with the ideals of R/M. So M is maximal if and only if there are no ideals I with $M \subset I \subset R$ if and only if there are no maximal ideals of R/M if and only if the only ideals of R/M are R/M and 0 if and only if R/M is a field. Some things to verify: R/M is commutative (since R is commutative, we have R/M is commutative), R/M has unity ($1 + M \in R/M$), and R/M is not the zero ring ($M \neq R, R/M \neq 0$). □

Example 1.6.1. What are the maximal ideals in $\mathbb{Z}$? We know all subrings of Z are of the form $n\mathbb{Z}, n \geq 0$. It takes very little work to show that $n\mathbb{Z}, n \geq 0$ is an ideal. The maximal ideals are those such that $\mathbb{Z}/n\mathbb{Z}$ is field. We know that $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime. Thus $n\mathbb{Z}, n \geq 0$ is a maximal ideal if and only if n is prime.

We now look to generalize the notion of a prime number through ideals. Looking at $\mathbb{Z}$, suppose $n\mathbb{Z}, n \geq 0$ is an ideal of $\mathbb{Z}$. Suppose $a, b \in \mathbb{Z}$ such that $a \in n\mathbb{Z}$ or $b \in n\mathbb{Z}$. Note

$$a \in (n) \iff a = nx \quad \text{for some integer } x$$

So $n \mid a$. Equivalently, we can say if $n \mid ab$ then $n \mid a$ or $n \mid b$. This is exactly the behavior of a prime number. This motivates the following definition.

Definition 1.6.2. (Prime Ideal)
Assume R is commutative. An ideal P is called a prime ideal if $P \neq R$ and if $ab \in P$, then $a \in P$ or $b \in P$.

The defining trait of a prime ideal is that whenever a product lands in the ideal, one of the factors must already be there. In a quotient ring, this looks like the following: if $(a + P)(b + P) = 0 + P$ then either $(a + P) = 0 + P$ or $(b + P) = 0 + P$. This leads us to the following characterization.

Proposition 1.6.4. Assume R is commutative. The ideal P is a prime ideal in R if and only if R/P is an integral domain.

Proof. The ideal P is prime if and only if $P \neq R$ and if $ab \in P$, then $a \in P$ or $b \in P$. Recall that

(i) $ab \in P \iff (a + P)(b + P) = ab + P = 0 + P$

(ii) $a \in P \iff a + P = 0 + P$

(iii) $b \in P \iff b + P = 0 + P$

So, P is a prime ideal if and only if $R/P \neq 0$ and if $(a + P)(b + P) = 0 + P$, then $a + P = 0$ or $b + P = 0$ if and only if R/P is an integral domain (since we know R/P is a commutative ring with unity). $\square$

1.7 Rings of Fractions

We take a brief look at fractions. Fractions are equivalence cleases, as given any two fractions a/b and c/d we have

$$\frac{a}{b} = \frac{c}{d} \iff da = cb$$

If we look at $R \times R$ where R is a ring we can define fractions as ordered pairs under the relation

$$(a, b) = (c, d) \iff da = cb$$

What this allows us to do when R is an integral domain is to create a field which contains an isomorphic copy of R. We will not spend much time here, so we end this chapter with the following remark.

Remark 1.7.1. Every integral domain is contained in a field and the smalles field which contains an integral domain is called the field of fractions of the integral domain.

In the next chapter, we will discuss different types of integral domains. More specifically, we will discuss the following:

(i) Euclidean domain (ED)

(ii) Principal ideal domain (PID)

(iii) Unique factorization domains (UFD)

Chapter 2

Euclidean Domains, Principal Ideal Domains, and Unique Factorization Domains

2.1 Euclidean Domains

Having seen how prime ideals generalize the idea of prime numbers to ring structures, and their relation to integral domains, we now focus more closely on the arithmetic of integral domains. In number theory, division with remainders and unique factorization are tightly linked concepts. We wish to generalize these concepts to integral domains, capturing when an integral domain supports a robust theory of greatest common divisors and factorizations.

We first start with integral domains that have a sense of a division algorithm. We will call these Euclidean Domains. First, here are some examples where a division algorithm exists.

Example 2.1.1. (i) $\mathbb{Z}$ is a euclidean domain.

(ii) $F[x]$, the set of polynomials over a field, is a euclidean domain.

(iii) Any field will be a euclidean domain.

(iv) The Gaussian integers, $\{a + bi : a, b \in \mathbb{Z}\}$, is a euclidean domain.

Before we formally define a euclidean domain, we first need to define special functions on a ring that would allow us to terminate an algorithm. More specifically, if we wish to have a division algorithm, we would need a way to specifiy that the remainder becomes provably smaller. This avoids the issue of having repeated remainders that continue forever.

Definition 2.1.1. (Norm on a Ring)
Any function $N : R \to \mathbb{N} \cup \{0\}$ (where R is a ring) with $N(0) = 0$ is called a norm on the ring R. If $N(a) > 0$ for all $a \in R, a \neq 0$, we call N a positive norm.

We now define euclidean domains.

Definition 2.1.2. (Euclidean Domain)
An integral domain R is said to be a euclidean domain (ED) if there exists a norm on R such that for any $a, b \in R$ with $b \neq 0$ there exists $q, r \in R$ such that

$$a = bq + r$$

where $r = 0$ or $N(r) < N(b)$.

A division algorithm gives us a euclidean algorithm. This algorithm is repeated division of some a and b in

the euclidean domain. Given a euclidean domain R and $a, b \in R$ with $b \neq 0$,

$$\begin{aligned} a &= q_0 b + r_0 \text{ where } r_0 = 0 \text{ or } N(r_0) < N(b) \\ b &= q_1 r_0 + r_1 \text{ where } r_1 = 0 \text{ or } N(r_1) < N(r_0) \\ r_0 &= q_2 r_1 + r_2 \text{ where } r_2 = 0 \text{ or } N(r_2) < N(r_1) \\ &\vdots \\ r_{n-2} &= q_n r_{n-1} + r_n \text{ where } r_n = 0 \text{ or } N(r_n) < N(r_{n-1}) \end{aligned}$$

Because the codomain of the norm is $\mathbb{N} \cup \{0\}$ we have created a strictly decreasing sequence so it must terminate with some remainder of zero.

$$N(b) > N(r_0) > N(r_1) > ... > N(r_{n-1}) > N(r_n)$$

Example 2.1.2. Some examples of norms.

(i) $\mathbb{Z}$, $N(x) = |x|$

(ii) $F[x]$, $N(f(x)) = \deg(f(x))$

(iii) For any field $\mathbb{F}$, any norm works: $N(x) = 0 \ \forall x \in \mathbb{F}$, $N(x) = 69 \ \forall x \in \mathbb{F}$, etc. There are no remainders when dividing in a field, so there is no need for a norm to terminate the euclidean algorithm.

(iv) $\mathbb{G} = \{a + bi : a, b \in \mathbb{Z}\}$ (the Gaussian integers), $N(a + bi) = a^2 + b^2$

Proposition 2.1.1. Every ideal in a euclidean domain is a principal ideal. More precisely, if I is any nonzero ideal in the euclidean domain R then $I = (d)$ where d is any nonzero element of I with minimum norm.

Proof. If I is the zero ideal, then $I = (0)$. Suppose $I \neq 0$. Then let d be a nonzero element of I with minimal norm (such a d exists since the set $\{N(a) : A \in I\}$ is a nonempty set of integers bounded below and by the well-ordering principal there must be a minimal norm attained by an element of I). Clearly, $(d) \subseteq I$ since $d \in I$. To show $I \subseteq (d)$ let $a \in I$ and use the division algorithm to write

$$a = qd + r$$

where $q, r \in R$ and $r = 0$ or $N(r) < N(d)$. Then

$$r = a - qd$$

and since I is an ideal and $q \in R$ with $d \in I$, we have $qd \in I$. Further, $a \in I$ and so $a - qd \in I$. Thus $r \in I$. Note that $N(d) < N(r)$, which contradicts minimality of the norm of d. Hence, the only option is that $r = 0$. Hence

$$a = qd \implies a \in (d)$$

Therefore, $I \subseteq (d)$. It follows that $I = (d)$. $\square$

2.2 Number Theory in a Ring

Now that we have a notion of a euclidean domains, we proceed to establish arithmetic on such rings, notably divisibility and greatest common divisors.

Definition 2.2.1. (Divide, Greatest Common Divisor)
Let R be a commutative ring and let $a, b \in R$ with $b \neq 0$. The element b is said to divide a if

$$\exists r \in R \text{ such that } br = a$$

If b divides a, we write $b \mid a$.
The greatest common divisor (GCD) of a and b is a nonzero element of R, d, such that

(i) $d \mid a$ and $d \mid b$

(ii) If $d' \mid a$ and $d' \mid b$, then $d' \mid d$

The notation remains the same:

$$d = \gcd(a, b) \text{ or } d = (a, b)$$

In the language of ideals if I is the ideal generated by (a, b) then d is a $\gcd(a, b)$ if

(i) $I \subseteq (d)$

(ii) If $I \subseteq (d')$ then $(d) \subseteq (d')$

Proposition 2.2.1. If a and b are nonzero elements of a commutative ring R such that the ideal generated by a and b is a principal ideal (d), then d is a greatest common divisor of a and b.

Proof. Suppose $I = (a, b) = (d)$. Then

$$a, b \in (d) \implies d \mid a \text{ and } d \mid b$$

Let $d' \in R$ such that $(a, b) \subseteq (d')$. Then $a \in (d')$ and $b \in (d')$ implies $d' \mid a$ and $d' \mid b$. Since $(d) = (a, b)$, $(d) \subseteq (d')$ which implies $d \in (d')$. More specifically, $d' \mid d$. Hence, d is a greatest common divisor of a and b. □

Just like for standard integers, the greatest common divisor is unique.

Proposition 2.2.2. Let R be an integral domain. If d and d' in R both generate the same principal ideal, i.e. $(d) = (d')$, then $d = ud'$ where u is a unit in R. In particular, if d and d' are both greatest common divisors of a and b, then $d' = wd$ where w is some unit in R.

Proof. This is trivial if d or d' is zero. Assume d and d' are both nonzero. Since $(d) = (d')$, we have that $d' \in (d)$ and so $\exists x \in R$ such that $dx = d'$. Since $d \in (d')$, $\exists y \in R$ such that $d'y = d$. This gives

$$\begin{aligned} d'yx = d' &\implies d'yx - d' = 0 \\ &\implies d'(yx - 1) = 0 \end{aligned}$$

Since $d' \neq 0$ and R is an integral domain, we know that $yx - 1 = 0 \implies yx = 1$. Hence, both x and y are units. □

2.3 Principal Ideal Domains

Previously, we have shown that every ideal in a euclidean domain is principal. Euclidean domains form a special case of what are called principal ideal domains. We now work in the more general principal ideal setting; in particular, each result for princiapl ideal domains that we prove will hold in any euclidean domain.

Definition 2.3.1. (Principal Ideal Domain)
A principal ideal domain (PID) is an integral domain where every ideal is principal.

Proposition 2.3.1. Let R be a principal ideal domain and let $a, b \in R$ where both a and b are nonzero. Let d be the generator of the ideal (a, b). Then

(i) d is a greatest common divisor of a and b

(ii) d can be written as an R linear combination of a and b: $\exists x, y \in R$ such that $ax + by = d$

(iii) d is unique up to unit multiples

In a commutative ring,

$$(d) = (a, b) = (a) + (b) = \{ax + by : x, y \in R\} \implies \exists x, y \in R \text{ such that } ax + by = d$$

Proposition 2.3.2. Every nonzero prime ideal in a principal ideal domain is maximal.

Proof. Let (p) be a nonzero prime ideal of a principal ideal domain R. Let I be an ideal that contains (p). Since R is a principal ideal domain, there exists some $i \in R$ such that $(i) = I$. Our goal is to show $I = (p)$ or $I = R$. Since $p \in (i)$, we have

$$p = ri \text{ for some } r \in R$$

So $ri \in (p)$. Since (p) is prime, $r \in (p)$ or $i \in (p)$. If $i \in (p)$ then $(i) \subseteq (p)$. Since $I = (i)$ was assumed to contain (p), we have $I = (p)$.
If $r \in (p)$, then $r = px$ for some $x \in R$. Then

$$p = ri \implies r = (ri)x$$

Since r is nonzero (ir $r = 0$ then $p = 0$, a contradiction), we have

$$1 = ix$$

Thus i and x are units. Since $i \in I$ we have that $I = R$. □

Fact 2.3.1. Here are some facts about $R[x]$ where R is a commutative ring:

(i) If R is an integral domain, $R[x]$ is an integral domain

(ii) R is contained in $R[x]$ where R is the set of all constant polynomials

Proposition 2.3.3. If R is any commutative ring such that the polynomial ring $R[x]$ is a principal ideal domain, then R is a field.

Proof. Assume $R[x]$ is a principal ideal domain. Since R is a subring of $R[x]$, then R must be an integral domain. The ideal (x) is a nonzero prime ideal in $R[x]$ because

$$\frac{R[x]}{(x)} \cong R \tag{$*$}$$

is an ideal. Thus (x) is maximal, so $R[x]/(x)$ is a field. That is, R is a field.

Proof of $(*)$: Let $\sum_{i=0}^{n} a_i x^i \in R[x]$. Then in $R[x]/(x)$, we have

$$\sum_{i=0}^{n} a_i x^i + (x) = a_0 + (x)$$

Let's define a mapping

$$\psi : R[x] \to R$$
$$\psi(f) = f(0)$$

Let $f(x), g(x) \in R[x]$ where $f(x)$ has constant coefficient a_0 and $g(x)$ has constant coefficient b_0. Then

$$\psi(f(x)g(x)) = \psi(h(x)) = h(0) = a_0 b_0 = f(0)g(0) = \psi(f(x))\psi(g(x))$$

where $h(x) = f(x)g(x)$. Similarly,

$$\psi(f(x) + g(x)) = \psi(k(x)) = k(0) = a_0 + b_0 = f(0) + g(0) = \psi(f(x)) + \psi(g(x))$$

where $k(x) = f(x) + g(x)$. If $f(x) \in \ker \psi$, the $f(0) = 0$. That is, $f(x)$ has a zero constant coefficient. That is,

$$f(x) = a_1 x + a_2 x^2 + ... + a_n x^n = x(a_1 + a_2 x + ... + a_n x^{n-1})$$

so $f(x) \in (x)$. Note that if $p(x) \in (x)$, then $p(x)$ has zero constant coefficient so $p(0) = 0$ and $p(x) \in \ker \psi$. By the first isomorphism theorem,

$$\frac{R[x]}{\ker \psi} \cong \psi(R[x])$$
$$\implies \frac{R[x]}{(x)} \cong \psi(R[x])$$

Note ψ is onto since for any $r \in R$, $\psi(r) = r$ where we consider $r \in R[x]$ to be a constant polynomial. Hence

$$\frac{R[x]}{(x)} \cong R$$

□

2.4 Unique Factorization Domains

When finding the greatest common divisor for integers, we use the fundamental theorem of arithmetic to find prime factorizations. For example, if we have $m = 3^7 \cdot 2^3 \cdot 5^6$ and $n = 7^7 \cdot 2^5 \cdot 3^2$ then the greatest commond divisor is

$$\gcd(m, n) = 3^{\min\{2,7\}} \cdot 2^{\min\{3,5\}} = 3^2 \cdot 2^3$$

We would to extend this mechanism to rings.

Definition 2.4.1. (Reducible, Prime, Associate)
Let R be an integral domain.

(i) Suppose $r \in R$ is nonzero and not a unit. Then r is called irreducible if whenever $r = ab$ where $a, b \in R$, then one of a and b is a unit. Otherwise, r is said to be reducible.

(ii) A nonzero element $p \in R$ is called prime if (p) is a prime ideal (if $ab \in (p)$, then $a \in (p)$ or $b \in (p)$).

(iii) Two elements a and b of R differing by a unit multiple are called associate in R (i.e., $a = ub$ where $u \in R$ is a unit).

Proposition 2.4.1. In an integral domain, a prime is irreducible.

Proof. Suppose p is a prime in an integral domain R. So (p) is a nonzero prime ideal. Let $p = ab$ where $a, b \in R$. Then $ab \in (p)$. Since (p) is prime, $a \in (p)$ or $b \in (p)$. Without loss of generality, suppose $a \in (p)$. Then $a = pr$ where $r \in R$. Then

$$\begin{aligned} & p = (pr)b \\ \implies & 1 = rb \\ \implies & b \text{ is a unit in } R. \end{aligned}$$

Hence p is irreducible in R. □

In an integral domain, a prime element is irreducible. However, not every irreducible element is prime. For example $3 \in \left\{a + b\sqrt{-5} : a, b \in \mathbb{Z}\right\}$ is irreducible but not prime. Now if we consider a principal ideal domain, then we have the following result.

Proposition 2.4.2. In a principal ideal domain, a nonzero element is prime if and only if it is irreducible.

Proof. Let R be a principal ideal domain. Since principal ideal domains are integral domains, it automatically follows that any prime $p \in R$ is irreducible in R.
Let $p \in R$ be irreducible. Consider the ideal generated (p). Let M be an ideal such that $(p) \subseteq M$. Since R is a principal ideal domain, there exists $m \in R$ such that $M = (m)$. So $p \in (m)$ and thus $p = mx$ for some $x \in R$. Since p is irreducible, x or m must be a unit. If x is a unit, then m and p are associate, hence $(m) = (p)$. If m is a unit, then $(m) = R$. Hence (p) is maximal, which implies (p) is prime. □

Principal ideal domains provide a nice characterization for a prime involving reducibility. This mirrors how primes behave in the integers: a prime $p \in \mathbb{N}$, $p > 1$ is prime if its only factors are 1 and p (p is irreducible in $\mathbb{N}$). We now classify rings where factorizations are unique, allowing us to have a notion of the fundamental theorem of arithmetic.

Definition 2.4.2. (Unique Factorization Domain)
A unique factorization domain (UFD) is an integral domain R in which every nonzero $r \in R$ that is not a unit has the following properties:

(i) r can be written as a finite product of irreducibles p_i of R (not necessarily distinct)

$$r = p_1 p_2 \cdots p_n$$

(ii) The decomposition in part (i) is unique up to associates, namely if $r = q_1 q_2 \cdots q_m$ is another factorization of r, then $m = n$ and there is some reording of the factors such that $p_i = u_i q_i$ for all i where u_i is a unit in R.

Proposition 2.4.3. In a unique factorization domain a nonzero element is prime if and only if it is irreducible.

Proof. We only need to prove ($\Longleftarrow$). To that end, let p be an irreducible element in R and assume $p \mid ab$ for some $a, b \in R$. We will show that $p \mid a$ or $p \mid b$. Since $p \mid ab$, there exists some $d \in R$ such that $pd = ab$. Since R is a unique factorization domain, a and b can be written uniquely (up to assoicates) a product of irreducibles in R. That is,

$$a = p_1 p_2 \cdots p_n$$
$$b = q_1 q_2 \cdots q_m$$

where the p_i and q_i are irreducibles. Hence

$$ab = (p_1 p_2 \cdots p_n)(q_1 q_2 \cdots q_m)$$

which gives a unique (up to associates) factorization of ab. Thus

$$pd = p_1 p_2 \cdots p_n q_1 q_2 \cdots q_m$$

By the uniqueness of factorizations we know that p is an associate of one of the irreducibles on the right-hand side. If p is an associate of a p_i for some i, then $p \mid a$. Similarly, if p is an associate of a q_j for some j, then $p \mid b$. Thus p is prime. □

At this point, we have introduced three types of integral domains. We conclude this section by showing that every principal ideal domain is in fact a unique factorization domain. First, we need the following lemma.

Lemma 2.4.1. In a principal ideal domain R, given an ascending chain of ideals

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq ... \subseteq R$$

it eventually becomes stationary. That is,

$$\exists n \in \mathbb{N} \text{ such that } I_n = I_k \forall \ \ k \geq n$$

Proof. Let $I = \bigcup_{i=1}^{\infty} I_i$. It is easy to show I is an ideal. Since R is a principal ideal domain,

$$\exists p \in R \text{ such that } I = (p)$$

Since $p \in I = \bigcup_{i=1}^{\infty} I_i$, $p \in I_n$ for some $n \in \mathbb{N}$. Thus $(p) \subseteq I_n$ and by definition of unions $I_n \subseteq I = (p)$. Thus $I_n = (p)$ and for any $k \geq n$, we have that $I_k \supseteq I_n \supseteq I$ so it follows that $I_k = I$. □

Theorem 2.4.1. Every principal ideal domain is a unique factorization domain.

Proof. Let $r \in R$ where R is a principal ideal domain. Further assume that r is nonzero and not a unit. If r is irreducible, then we're done. If not, r is reducible so

$$r = r_1 r_2 \text{ for } r_1, r_2 \in R$$

where r_1 and r_2 are not units. If both r_1 and r_2 are reducible, then we're done. Otherwise, r_1 and r_2 are reducible. Supposing r_1 is reducible,

$$r_1 = r_{11} r_{12} \text{ for } r_{11}, r_{12} \in R$$

and so forth. If this process does not terminate, we obtain a proper inclusion of ideals. That is,

$$(r) \subset (r_1) \subset (r_{11}) \subset ...$$

where (r) is prperly in (r_1) since r_2 is not a unit and hence r and r_1 are associates. The rest of the proper containment follows similarly. However, in a principal ideal domain any ascending chain of ideals become stable so this contradicts the proper containment. Hence, we can factor r into a product of irreducible elements of R. □

To sumarize this chapter: every euclidean domain is a principal ideal domain, and every principal ideal domain

is a unique factorization domain.

$$\text{Euclidean domains} \subseteq \text{principal ideal domains} \subseteq \text{unique factorization domains}$$

Chapter 3

Polynomials and Vector Spaces

3.1 Polynomials Over a Field

Now that we have a foundation for factorization in a ring setting, we shift our focus to polynomial rings.

Recall . Let R be an integral domain. Then

(i) $\forall p, q \in R[x]$ with $p \neq 0$ and $q \neq 0$, $\deg(p(x)q(x)) = \deg(p(x)) + \deg(q(x))$.

(ii) The units in $R[x]$ are just units of R.

(iii) $R[x]$ is an integral domain.

If $R[x]$ is an integral domain, the field of fractions of $R[x]$ is

$$R(x) = \left\{ \frac{p(x)}{q(x)} : p, q \in R[x], q(x) \neq 0 \right\}$$

Proposition 3.1.1. Let I be an ideal of R and let $(I) = I[x]$ (the ideal generated by I in $R[x]$). Then

$$\frac{R[x]}{(I)} \cong \left(\frac{R}{I}\right)[x]$$

In particular, if I is a prime ideal of R, then this is a prime ideal of $R[x]$.

Proof. Define $\varphi : R[x] \to (R/I)[x]$ such that

$$\sum_{i=0}^{n} a_i x^i \mapsto \sum_{i=0}^{n} \bar{a_i} x^i$$

where $\bar{a_i} = a_i + I$. By the definition of multiplication and addition of cosets, φ is a ring homomorphism. Let $p(x) \in R[x]$ such that

$$p(x) = \sum_{i=0}^{n} a_i x^i$$
$$\bar{p(x)} = \sum_{i=0}^{n} \bar{a_i} x^i$$

We know

$$\begin{aligned} p(x) \in \ker\varphi &\iff \varphi(p(x)) = \bar{0} \\ &\iff \bar{p(x)} = \bar{0} \\ &\iff \bar{a_i} = \bar{0} \quad \forall i \\ &\iff a_i \in I \quad \forall i \end{aligned}$$

Thus $\ker\varphi = I[x] = (I)$. Hence the result follows from the first isomorphism theorem.
The second statement could be shown by letting I be a prime ideal in R. It follows that R/I is an integral

domain. Hence $(R/I)[x]$ is an integral domain. Since

$$\left(\frac{R}{I}\right)[x] \cong \frac{R[x]}{(I)}$$

we have that $R[x]/(I)$ is an integral domain. We conclude (I) is a prime ideal. □

Example 3.1.1. Consider $R = \mathbb{Z}$ and $I = n\mathbb{Z}$. Then

$$\frac{\mathbb{Z}[x]}{n\mathbb{Z}[x]} \cong \left(\frac{\mathbb{Z}}{n\mathbb{Z}}\right)[x]$$

If n is composite, then $(\mathbb{Z}/n\mathbb{Z})[x]$ is not an integral domain. Hence, $\mathbb{Z}[x]/n\mathbb{Z}[x]$ is not an integral domain. So $n\mathbb{Z}[x]$ is not prime. If n is a prime p, then

$$\left(\frac{\mathbb{Z}}{p\mathbb{Z}}\right)[x] \text{ is an integral domain.}$$

Hence, $\mathbb{Z}[x]/p\mathbb{Z}[x]$ is an integral domain, and so $p\mathbb{Z}[x]$ is a prime ideal of $\mathbb{Z}[x]$.

We now focus our attention to polynomials over a field. Let F be a field and define a norm on $F[x]$ by $N(p(x)) = deg(p(x)) \quad \forall p \in F[x]$.

Theorem 3.1.1. (Polynomails Over Fields are Euclidean Domains)
Let F be a field. Then $F[x]$ is a euclidean domain. Specifically, if $a, b \in F[x]$ such that $b(x) \neq 0$ then there exists unique $q, r \in F[x]$ such that

$$a(x) = q(x)b(x) + r(x)$$

where $r(x) = 0$ or $\deg(r) < \deg(b)$.

Proof. (Existence) We proceed by induction. If $a(x) = 0$, the $q(x) = 0$ and $r(x) = 0$ gives us what we want. If $a(x)$ is a nonzero constant, then we take $q(x) = 0$ and $r(x) = a(x)$ if $\deg(b) > 0$. We have $a(x) = 0 \cdot b(x) + a(x)$. If $b(x)$ is a nonzero constant then we write $b(x) = b$ and take $q(x) = a(x)/b$, $r(x) = 0$. Then $a(x) = q(x)b(x) + r(x)$.
Let $\deg(a(x)) = n$ and $\deg(b(x)) = \mathrm{m}$, and suppose the result holds for all polynomials of degree less than n. If $n < m$, we let $q(x) = 0$ and $a(x) = r(x)$. If $n \geq m$, let

$$a(x) = \sum_{i=0}^{n} a_i x^i = a_n x^n + ... + a_0$$
$$b(x) = \sum_{i=0}^{m} b_i x^i = b_m x^m + ... + b_0$$

where b_m and a_n are nonzero. Then

$$a'(x) = a(x) - \frac{a_n}{b_m} x^{n-m} \cdot b(x)$$

is a polynomial with $\deg(a') < \deg(a)$. By our inductive hypothesis

$$a'(x) = q'(x)b(x) + r(x)$$

for some $q'(x), r(x) \in F[x]$ where $r(x) = 0$ or $\deg(r) < \deg(b)$. Letting

$$q(x) = q'(x) + \frac{a_n}{b_m} x^{n-m}$$

we have

$$\begin{aligned} q(x)b(x) + r(x) &= \left(q'(x) + \frac{a_n}{b_m} x^{n-m}\right) b(x) + r(x) \\ &= q'(x)b(x) + \frac{a_n}{b_m} x^{n-m} b(x) + r(x) \\ &= a'(x) + \frac{a_n}{b_m} x^{n-m} b(x) \\ &= a(x) \end{aligned}$$

completing the induction.
(Uniqueness) Suppose $q_1(x)$ and $r_1(x)$ also satisfy the conditions of the theorem. Then

$$\begin{aligned} q(x)b(x) + r(x) &= q_1(x)b(x) + r_1(x) \\ \implies b(x)(q(x) - q_1(x)) &= r_1(x) - r(x) \end{aligned}$$

If $q(x) - q_1(x) \neq 0$, then

$$\deg[b(x)(q(x) - q_1(x))] = \deg(b(x)) + \deg(q(x) - q_1(x)) \geq m$$

However, $\deg(r_1(x) - r(x)) < m$. Hence,

$$\begin{aligned} q(x) - q_1(x) &= 0 \\ \implies q(x) &= q_1(x) \end{aligned}$$

It follows immediately that $r(x) = r_1(x)$. □

Corollary 3.1.1. If F is a field, then $F[x]$ is a principal ideal domain and a unique factorization domain.

This result follows from the fact that all euclidean domains are principal ideal domains, and all principal ideal domains are unique factorization domains.

Example 3.1.2. $\mathbb{Q}[x]$ is a principal ideal domain and a unique factorization domain.

Example 3.1.3. $\mathbb{Z}[x]$ is not a principal ideal domain. Examine the ideal $(2, x)$ in $\mathbb{Z}[x]$. Recall that

$$(2, x) = \{2p(x) + xq(x) : p(x), q(x) \in \mathbb{Z}[x]\}$$

These are polynomials with even constant coefficients.

$$(2, x) \neq \mathbb{Z}[x]$$

since $x + 1 \in \mathbb{Z}[x]$ but not in $(2, x)$.
Suppose there exists $d(x) \in \mathbb{Z}[x]$ such that $(d(x)) = (2, x)$. So $2 \in (d(x))$, hence there exists $f(x) \in \mathbb{Z}[x]$ such that

$$d(x)f(x) = 2 \text{ with } \deg(2) = 0$$

so $\deg(d(x)) = 0 = \deg(f(x))$ and

$$d(x) \in \{\pm 1, \pm 2\}$$

3.2 Irreducibility Criterion

Now that we can discuss the factorizations of polynomials over fields, we would look to discuss the irreducibility of these polynomials. Take a look at this example in the rationals.

Example 3.2.1. In $\mathbb{Q}[x]$, $7x$ is irreducible. We can only factor

$$7x = a(x)b(x)$$

where either $a(x)$ is a unit or $b(x)$ is a unit. Suppose $a(x)$ and $b(x)$ are nonunits. The units of $\mathbb{Q}[x]$ are exactly the units in $\mathbb{Q}$ so if $a(x)$ and $b(x)$ are nonunits they must have positive degree which implies

$$\deg(7x) > 2$$

a contradiction. However, $7x$ is reducible in $\mathbb{Z}[x]$: let $a(x) = 7$ and $b(x) = x$. We know that every intregral domain is contained in its field of fractions, for instance

$$\mathbb{Z} \subseteq \mathbb{Q} \implies \mathbb{Z}[x] \subseteq \mathbb{Q}[x]$$

We will take a look at theorems which relate irreducibility in $\mathbb{Q}[x]$ to irreducibility in $\mathbb{Z}[x]$, or more generally, factorization of polynomials in $R[x]$ with factorization in $F[x]$ (where F is the field of fractions of the integral domain R).

Lemma 3.2.1. (Gauss's Lemma Generalized)
Let R be a unique factorization domain with field of fractions F and let $p(x) \in R[x]$. If $p(x)$ is reducible in $F[x]$, then $p(x)$ is reducible in $R[x]$.

Lemma 3.2.2. (Gauss's Lemma Special)
Let $p(x) \in \mathbb{Z}[x]$. If $p(x)$ is reducible in $\mathbb{Q}[x]$, then $p(x)$ is reducible in $\mathbb{Z}[x]$.

Corollary 3.2.1. Let R be a unique factorization domain with field of fractions F. Let $p(x) \in R[x]$ and suppose all the coefficients of $p(x)$ are relatively prime. Then

$$p(x) \text{ is irreducible in } R[x] \iff p(x) \text{ is irreducible in } F[x]$$

We will work with polynomials with coefficients in a field F. Hence $F[x]$ is a euclidean domain, a principal ideal domain, and a unique factorization domain. The irreducible polynomials in $F[x]$ are all non-constants. In particular, all linear polynomials in $F[x]$ are irreducible.

Proposition 3.2.1. Let F be a field with $p(x) \in F[x]$. Then $p(x)$ has a linear factor if and only if $p(x)$ has a root in F.

Proof. If $p(x)$ has a linear factor, we may assume it is of the form $x - \alpha$ for some $\alpha \in F$. But then $p(x) = (x - \alpha)g(x)$ for some $g(x) \in F[x]$. It follows

$$p(\alpha) = (\alpha - \alpha)g(\alpha) = 0$$

Conversely, suppose α is a root of $p(x)$. Then by the division algorithm,

$$\exists q(x), r(x) \in F[x] \text{ such that } p(x) = q(x)(x - \alpha) + r(x)$$

where $r(x) = 0$ or $\deg(r) < 1$. Hence $r(x)$ is a constant polynomial. Since α is a root of $p(x)$,

$$p(\alpha) = (\alpha - \alpha)q(\alpha) + r(\alpha) = r(\alpha)$$

Since $r(x)$ is a constant, $r(x) = 0$. That is, $p(x)$ has a linear factor. □

Proposition 3.2.2. A polynomial of degree 2 or 3 is reducible if and only if it has a root in F.

We will need tools to find roots of polynomials.

Theorem 3.2.1. (Rational Root Theorem)
Let $p(x) = a_n x^n + ... + a_0 \in Z[x]$ where $\deg(p) = n$. If $n/s \in \mathbb{Q}$ with $\gcd(r, s) = 1$ and $p(r/s) = 0$, then $r \mid a_0$ and $s \mid a_n$. In particular, if $p(x)$ is monic ($a_n = 1$) in $\mathbb{Z}[x]$ and $p(d) \neq 0$ for all d such that $d \mid a_0$, then $p(x)$ has no roots in $\mathbb{Q}$.

Proof. Assume $p(r/s) = 0 = a_n(r/s)^n + ... + a_1(r/s) + a_0$. Multiplying through by s^n :

$$\begin{aligned} 0 &= a_n r^n + a_{n-1} r^{n-1} s + ... + a_1 r s^{n-1} + a_0 s^n \\ \implies a_0 s^n &= -a_n r^n - a_{n-1} r^{n-1} s - ... - a_1 r s^{n-1} \\ &= r(-a_n r^{n-1} - a_{n-1} r^{n-2} s - ... - a_1 s^{n-1}) \end{aligned}$$

Hence $r \mid a_0 s^n$, but $\gcd(r, s) = 1$ so $r \mid a_0$. Similarly,

$$\begin{aligned} a_n r^n &= -a_{n-1} r^{n-1} s - ... - a_1 r s^{n-1} - a_0 s^n \\ &= s(-a_{n-1} r^{n-1} - ... - a_1 r s^{n-2} - a_0 s^{n-1}) \end{aligned}$$

Hence $s \mid a_n r^n$, but $\gcd(r, s) = 1$ so $s \mid a_n$. □

Example 3.2.2. (i) $x^3 - 3x - 1$ is irreducible (over $\mathbb{Q}$) in $\mathbb{Q}[x]$. By the rational root theorem, a rational root would be ± 1, neither of which yield zero when plugged in.

(ii) $x^3 - 3x - 1$ in $(\mathbb{Z}/2\mathbb{Z})[x]$ is still irreducible.

(iii) $x^3 - 3x - 1$ in $(\mathbb{Z}/3\mathbb{Z})[x]$ is reducible since 1 is a root $\mod 3$.

Proposition 3.2.3. Let I be a prime ideal in the integral domain R. Let $p(x) \in R[x]$ be a non-constant polynomial with leading coefficient not in I. If $\bar{p(x)} \in (R/I)[x]$ cannot be factored in $(R/I)[x]$ into two polynomials of positive degree, then $p(x)$ cannot be factored similarly in $R[x]$.

Proof. Suppose $p(x)$ can be factored in $R[x]$. Then there exists $a(x), b(x) \in R[x]$ where $a(x)$ and $b(x)$ are non-constant polynomials in $R[x]$ such that $a(x)b(x) = p(x)$. Hence $\bar{a(x)} \cdot \bar{b(x)} = \bar{p(x)}$ and $\bar{a(x)}, \bar{b(x)}$ are nonconstant polynomials. Since the leading coefficient of $p(x)$ is not in I, the leading coefficients of $a(x)$ and $b(x)$ must also not be in I. Hence the leading coefficients are not $\bar{0}$ in R/I. □

Example 3.2.3. $p(x) = x^2 + x + 1 \in \mathbb{Z}[x]$. Examine $p(x)$ in $(\mathbb{Z}/2\mathbb{Z})[x]$ and notice $p(0) = 1$.

$$p(1) = 3 \equiv 1 \mod 2$$

Hence $p(x)$ is irreducible in $(\mathbb{Z}/2\mathbb{Z})[x] \implies p(x)$ is irreducible in $\mathbb{Z}[x] \implies p(x)$ is irreducible in $\mathbb{Q}[x]$.

Theorem 3.2.2. (Eisenstein's Criterion)
Let P be a prime ideal of an integral domain R. Let $f(x) = \sum_{i=0}^{n} a_i x^i \in R[x]$ be a polynomial with degree greater than 0. Suppose $a_0, ..., a_{n-1} \in P$, $a_n \notin P$, and $a_0 \in P^2$. Then $f(x)$ is irreducible in $R[x]$ ($P^2 = \{\sum_{i=1}^{n} x_i y_i : n \in \mathbb{Z}, x_i y_i \in P\}$).
In the integers: Let $p \in Z$ be prime and $f(x) = \sum_{i=0}^{n} a_i x^i \in \mathbb{Z}[x]$ with degree $n \geq 1$. Suppose $p \mid a_i \quad \forall i \in \{0, 1, ..., n-1\}$, but $p \nmid a_n$ and $p^2 \nmid a_0$. Then $f(x)$ is irreducible* in $\mathbb{Z}[x]$, hence it is irreducible in $\mathbb{Q}[x]$.
$*f(x)$ cannot be factored into a product of two polynomials of positive degree.

Proof. Suppose that $f(x)$ can be written as a product of two polynomials of positive degree in $\mathbb{Z}[x]$. That is, $f(x) = a(x)b(x)$, $a(x), b(x) \in \mathbb{Z}[x]$ with $\deg(a) > 0$ and $\deg(b) > 0$. Consider $\bar{f(x)}, \bar{a(x)}$, and $\bar{b(x)}$ to be the reductions of $f(x), a(x)$, and $b(x)$ in $(\mathbb{Z}/p\mathbb{Z})[x]$ respectively. Since p divides all the coefficients of f except a_n, we have that

$$\bar{f(x)} = \bar{a_n} x^n \text{ since } \bar{f(x)} = \bar{a(x)}\bar{b(x)} = \bar{a(x)}\bar{b(x)}$$

We have $\bar{a(x)}\bar{b(x)} = \bar{a_n} x^n$. Since $\mathbb{Z}/p\mathbb{Z}$ is a field, $(\mathbb{Z}/p\mathbb{Z})[x]$ is a euclidean domain and hence a principal ideal domain, and further a unique factorization domain. Since x is an irreducible element of $(\mathbb{Z}/p\mathbb{Z})[x]$, we know that $\bar{a_n} x^n$ is the unique factorization up to associates of $\bar{f(x)}$ in $(\mathbb{Z}/p\mathbb{Z})[x]$. Thus $\bar{a(x)} = \bar{c} x^{n-r}$ and $\bar{b(x)} = \bar{b} x^r$ for some $b, c \in \mathbb{Z}$ where $\bar{b}\bar{c} = \bar{a_n}$ and $r < n$. Note $r \neq 0$ as this would imply that $\bar{b(x)}$ is constant and $r \neq n$ which implies $\bar{a(x)}$ is constant. The equation $\bar{a(x)} x^n = (\bar{c} x^{n-r})(\bar{b} x^r)$ implies that x

divides

$$\bar{c}x^{n-r} \text{ or } \bar{b}x^r$$

since x is also a prime.
$\bar{a(x)}$ and $\bar{b(x)}$ are the reductions of $a(x)$ and $b(x)$ in $(\mathbb{Z}/p\mathbb{Z})[x]$. This implies that the constant coefficients of both $\bar{a(x)}$ and $\bar{b(x)}$ are $\bar{0}$. That is, the constant coefficient of $a(x)$ and $b(x)$ must be divisible by p. Since $f(x) = a(x)b(x)$, the constant coefficient of $f(x)$ is divisible by p^2, a contradiiction. □

Proposition 3.2.4. $F[x]/(p(x))$ is a field if and only if $p(x)$ is irreducible.

Example 3.2.4. (i) $x^4 + 10x + 5 \in \mathbb{Z}[x]$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ since $5 \mid 10 = a_2$, $5 \mid 5 = a_0$, $5 \nmid 1 = a_4$, and $5^2 \nmid 5 = a_0$.

(ii) Let $a \in \mathbb{Z}$ such that $p \mid a$ but $p^2 \nmid a$ for some prime $p \in \mathbb{Z}$. Then $x^n - a$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ by Eisenstein's criterion for all $n \in \mathbb{N}$. In particular, $x^n - p$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ for all $n \in \mathbb{N}$ and for all primes p. $x^3 - 7$ has no zeros in $\mathbb{Q}$, but has zeros outside of $\mathbb{Q}$:

$$(\sqrt[3]{7}, \sqrt[3]{7}w, \sqrt[3]{7}w^2) \text{ where } w = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

Example 3.2.5. $f(x) = x^4+1$ is irreducible over $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$, but Eisenstein's criterion does not apply... or does it? Let's examine:

$$g(x) = f(x+1) = (x+1)^4 + 1 = x^4 + 4x^3 + 6x^2 + 4x + 2$$

We see that $g(x)$ is irreducible by Eisenstein's criterion with $p = 2$. Suppose $f(x) = a(x)b(x)$ where $a(x), b(x) \in \mathbb{Z}[x]$ with positive degrees. Then

$$f(x) = a(x)b(x) \iff f(x+1) = a(x+1)b(x+1)$$

Example 3.2.6. (Important Example) Cyclotomic Polynomial. Let p be a prime in $\mathbb{Z}$. Consider

$$\phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \ldots + x + 1$$

Then

$$\begin{aligned}\phi_p(x+1) &= \frac{(x+1)^p - 1}{x+1-1} = \frac{\sum_{k=0}^{p} \binom{p}{k} x^k 1^{p-k} - 1}{x} \\ &= \frac{\sum_{k=1}^{p} \binom{p}{k} x^k}{x} \\ &= \sum_{k=1}^{p} \binom{p}{k} x^{k-1}\end{aligned}$$

Then $a_0 = k$, $a_n = 1$, and $p \mid a_k$ for all $0 \leq k < n$, $p \nmid a_n$, and $p^2 \mid a_0$. By Eisenstein's criterion, $\phi_p(x+1)$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$.

3.3 Vector Spaces

We now briefly take some time to discuss vector spaces. We will introduce some basic definitions and properties without going into too much detail. We are setting up the necessary background for field extensions.

Definition 3.3.1. (Vector Space)
Let F be a field. A vector space over F is F with a set V together with

(i) A binary operation $+$ on V such that $< V, + >$ is an abelian group.

(ii) An action of F on V (a mapping $f \times V$ to V), denoted kv for every $k \in F$ and $v \in V$, which satisfies

(a) $(j+k)v = jv + kv$ for all $j, k \in F, v \in V$
(b) $(jk)v = j(kv)$ for all $j, k \in F, v \in V$
(c) $j(v+w) = jv + jw$ for all $j \in F, v, w \in V$
(d) $1v = v$ for all $v \in V$

Definition 3.3.2. (Finite Linear Combination)
Given $v_1, ..., v_n \in V$ a vector space and $\alpha_1, ..., \alpha_n \in F$ a field, a sum of the form

$$\alpha_1 v_1 + ... + \alpha_n v_n$$

is a finite linear combination of $v_1, ..., v_n$.

Definition 3.3.3. (Span of a Set)
Given $S \subseteq V$, the span of S, denoted $\text{span}(S)$, is the set of all finite linear combinations (lc) of elements in S:

$$\text{span}(S) = \left\{ \sum_{i=1}^{n} a_i s_i : n \in \mathbb{N}, \alpha_i \in F, s_i \in S \right\}$$

Definition 3.3.4. (Linear Independence)
A subset $S \subseteq V$ is called a set of linearly independent (li) vectors if the equation

$$\alpha_1 v_1 + ... + \alpha_n v_n = 0 \text{ for all } \alpha_i \in F, v_i \in S$$

implies that $\alpha_1 = ... = \alpha_n = 0$.

Definition 3.3.5. (Basis)
A basis of a vector space V is an ordered set of linearly independent vectors B which span V. That is,

$$\text{span}(B) = V$$

Proposition 3.3.1. Assume that $A = \{v_1, ..., v_n\} \subseteq V$ is spans V, but every $S \subset A$ does not span V. Then A is a basis of V.

Proof. We know $\text{span}(A) = V$. It remains to check that $v_1, ..., v_n$ are linearly independent vectors. Suppose for the sake of contradiction

$$\alpha_1 v_1 + ... + \alpha_n v_n = 0$$

for $\alpha_i \in F$ and not all α_i are zero. Assume without loss of generality that $\alpha_1 \neq 0$. Then

$$v_1 = \frac{1}{\alpha_1}(-\alpha_2 v_2 - ... - \alpha_n v_n) \tag{$*$}$$

It follows that

$$\text{span}(v_2, ..., v_n) = V$$

since any linear combination of $v_1, ..., v_n$ can be written as a linear combination of $v_2, ..., v_n$ using $(*)$. Hence we have arrived at a contradiction. $\square$

Example 3.3.1. Let F be a field and $p(x) \in F[x]$ be an irreducible polynomial. From our homework, we know that $F[x]/(p(x))$ is a field. We were told that $F[x]/(p(x))$ is a vector space over F. You check the details (scalar multiplication $k(f(x)+g(x)) = kf(x)+kg(x)$). We also found out in our homework that

$$\operatorname{span}(\{\bar{1}, \bar{x}, \bar{x^2}, ..., \bar{x^{n-1}}\}) = \frac{F[x]}{(p(x))}$$

To prove the claim $\bar{1}, ..., \bar{x^{n-1}}$ forms a basis, we only need to show the linear independence of $\bar{1}, ..., \bar{x^{n-1}}$. To that end, consider

$$\alpha_0\bar{1} + ... + \alpha_{n-1}\bar{x^{n-1}} = \bar{0}$$

with not all $\alpha_i = 0$. Then

$$\begin{aligned} &\alpha_0 + \alpha_1 x + ... + \alpha_{n-1}x^{n-1} + (p(x)) = 0 + (p(x)) \\ \implies &\alpha_0 + \alpha_1 x + ... + \alpha_{n-1}x^{n-1} \in (p(x)) \\ \implies &p(x) \mid (\alpha_0 + \alpha_1 x + ... + \alpha_{n-1}x^{n-1}) \end{aligned}$$

However, $\deg(p(x)) = n$ but this is impossible uless $a_0 + a_1x + ... + a_{n-1}x^{n-1}$ is the zero polynomial. Hence $\alpha_0 = ... = \alpha_{n-1} = 0$ so $\bar{1}, ..., \bar{x^{n-1}}$ are linearly independent vectors.

Corollary 3.3.1. Assume the set $A \subseteq V$ spans V. Then A contains a basis of V.

Proof. If A is a basis, we are done. Suppose A is not a basis. Then there exists $A_1 \subset A$ such that $\operatorname{span}(A_1) = V$. If A_1 is a basis, we are done. Otherwise, there exists $A_2 \subset A_1$ such that $\operatorname{span}(A_2) = V$. Since our containments are proper and finite, this process must terminate. □

Theorem 3.3.1. (Replacement Theorem)
Suppose $A = \{a_1, ..., a_n\} \subseteq V$ is a basis for V containing n elements and $B = \{b_1, ..., b_m\} \subseteq V$ is a set of linearly independent vectors in V. Then there is an ordering of $a_1, ..., a_n$ such that for each $k \in \{1, ..., m\}$ the set $\{b_1, ..., b_k, a_{k+1}, ..., a_n\}$ is a basis of V. In particular, $m \leq n$.

Proof. We will use induction on k.
If $k = 0$, then there is nothing to do since A is assumed to be a basis of V. Suppose that $B_k = \{b_1, ..., b_k, a_{k+1}, ..., a_n\}$ is a basis for V. Then $\operatorname{span}(B_k) = V$, so

$$b_{k+1} = \beta_1 b_1 + ... + \beta_k b_k + \alpha_{k+1}a_{k+1} + ... + \alpha_n a_n$$

for some $\beta_i, \alpha_i \in F$. Note that not all α_j equal zero, otherwise b_{k+1} is linearly independent of $b_1, ..., b_k$, contradicting the linear independence of B. Without loss of generality, assume $\alpha_{k+1} \neq 0$. Then

$$a_{k+1} = \frac{1}{\alpha_{k+1}}\left(b_{k+1} - (\beta_1 b_1 + ... + \beta_k b_k + \alpha_{k+1}a_{k+1} + ... + \alpha_n a_n)\right)$$

Hence,

$$\operatorname{span}(b_1, ..., b_k, b_{k+1}, a_{k+2}, ..., a_n) = \operatorname{span}(b_1, ..., b_k, a_{k+1}, a_{k+2}, ..., a_n) = V$$

It remains to show that

$$B_{k+1} = \{b_1, ..., b_k, b_{k+1}, a_{k+2}, ..., a_n\}$$

is a linear independent set. If

$$\beta_1' b_1 + ... + \beta_{k+1}' b_{k+1} + \alpha_{k+2}' a_{k+2} + ... + \alpha_n' a_n = 0$$

then

$$\beta_1' b_1 + ... + \beta_{k+1}'(\beta_1 b_1 + ... + \beta_k b_k + \alpha_{k+1}a_{k+1} + ... + \alpha_n a_n) + \alpha_{k+2}' a_{k+2} + ... + \alpha_n' a_n = 0$$

Noting that the right hand side of this equation is in B_k, we know that all (simplified) coefficients must be 0 since B_k is assumed to be linearly independent. In particular, the coefficient of a_{k+1} is $\beta_{k+1}'\alpha_{k+1} = 0$ so $\beta_{k+1}' = 0$ or $\alpha_{k+1} = 0$ since fields are integral domains. Recall $\alpha_{k+1} \neq 0$, hence $\beta_{k+1}' = 0$. But then we have

$$\beta_1' b_1 + ... + \beta_k' b_k + \alpha_{k+2}' a_{k+2} + ... + \alpha_n' a_n = 0$$

Thus all $\beta_i', \alpha_j' \in F$ must be zero since $b_1, ..., b_k, a_{k+1}, ..., a_n$ are linearly independent by the inductive hypothesis. □

Corollary 3.3.2. (i) Suppose V is a finite basis with n elements. Any set of linearly independent vectors has at most n elements.

(ii) If V has some finite basis, then any two bases of V have the same number of elements.

Definition 3.3.6. (Dimension of a Vector Space)
If V is a vector space over F with a finite basis, then the cardinality of any basis is called the dimension of V, denoted $\dim V$.